



Citrullination in proteins of the polycomb repressor complex 1

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ABSTRACT

The Polycomb repressive complexes can be considered a paradigm of chromatin-based gene regulation in animals. Two of the proteins involved in such complexes are the Ring finger protein 1 (RING1) and the intrinsically disordered protein RING1A and YY1 binding protein (RYBP), which binds to RING1 in some of the Polycomb complexes. Peptidyl arginine iminohydrolases (PADIs) are enzymes involved in the post-translational modification of arginine to citrulline. PADI4, one of the five isoforms of PADI in humans, citrullinates RING1B and RYBP at specific arginines at their C-terminal (Arg231) and N-terminal (Arg53) regions, respectively. In this work, we studied *in vitro* and *in silico* whether isolated peptides from each protein, encompassing those wild-type arginines and their citrulline-modified counterparts, were capable of binding to PADI4. The isolated peptides, either in wild-type or modified form, were monomeric and disordered as indicated by far-UV CD and NMR. The affinity constants of peptides for PADI4, as measured by fluorescence, biolayer interferometry (BLI) and isothermal titration calorimetry (ITC), were in the low micromolar range (dissociation constants around 10–20 μ M), and the affinities were smaller for the citrullinated variants. Molecular docking simulations showed that the isolated peptides from each protein, either in wild-type form or modified variants, targeted the active site of PADI4, corresponding to the catalytic residue Cys645. These findings suggest that the isolated peptides were capable of binding to the enzyme with affinities similar to those of intact RING1B or RYBP, and that citrullination of the arginines diminished the affinity for the enzyme.

1. Introduction

Specific and unique gene expression programs are necessary for successful development of multicellular organisms. Deciphering these intricate routes is currently under way, and results so far have suggested that histone and chromatin post-translational modification (PTM)

contribute to modulate transcription and gene expression [1–3]. Polycomb systems comprise several arrays of multi-protein complexes, which are involved in: (i) catalyzing histone modification; and (ii) interactions with nucleosomes, supporting mainly transcription repression of several genes as well as, in a few cases, an increase in gene activation [4–9].

Abbreviations: BAEE, N- α -Benzoyl-L-arginine ethyl ester; BLI, biolayer interferometry; CD, circular dichroism; C-RING1B, C-terminal region of RING1B (residues 228–336 of the intact protein); COLDER, color developed reagent; DIPSI, decoupling in the presence of scalar interactions; DOSY, diffusion ordered spectroscopy; ITC, isothermal titration calorimetry; MW, molecular weight; NMR, nuclear magnetic resonance; NOE, nuclear Overhauser effect; NOESY, nuclear Overhauser effect spectroscopy; PADI, peptidyl arginine iminohydrolase; PCGF, Polycomb group ring finger; PRC, Polycomb repressive complex; PTM, post-translational modification; RAWUL, RING-finger and WD40-associated ubiquitin-like; RING, really interesting new gene; RYBP, RING 1A and YY1 binding protein; TCEP, tris(2-carboxyethyl) phosphine; TOCSY, total correlation spectroscopy; TSC, thiosemicarbazide; TSP, 3-(trimethylsilyl) propionic acid-2,2,3,3- 2 H₄-sodium salt; UV, ultraviolet.

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Polycomb genes were firstly discovered over 70 years ago in *Drosophila melanogaster*, where they are needed for repression of the homeotic (Hox) genes, which are involved in body plan specification [10]. The mammalian orthologues are crucial in modulating gene expression during development [11]. Polycomb proteins assemble into one of two large multi-protein complexes capable of modifying histones. The Polycomb repressive complex 1 (PRC1) is an E3 ubiquitin ligase capable of mono-ubiquitylating histone H2A at Lys119 [12–15] (yielding H2AK119ub1). On the other hand, the Polycomb repressive complex 2 (PRC2) is capable of mono-, di- and tri-methylate histone H3 at Lys27 [16,17]. Both PRC1 and PRC2 form the so-called Polycomb chromatin domains, which are formed by chromatin and specially enriched in H2AK119ub1, H3K27me3 and the Polycomb complexes themselves [18,19].

The catalytic core of the E3-ubiquitin ligase PRC1 is formed by RING1B (UniProt code: Q99496) – or alternatively its paralogue RING1A – and one of the six Polycomb group ring finger (PCGF) proteins. The RING1 and PCGF proteins have a similar domain arrangement: an N-terminal RING domain and a C-terminal RING-finger and WD40-associated ubiquitin-like (RAWUL) domain [20]. To ubiquitinate the nucleosomes, RING1 and PCGF proteins hetero-dimerize through their RING domains, allowing interaction with an E2 conjugating enzyme [20–23]. The RAWUL domain binds to a range of other auxiliary proteins, which are selected by the PCGF components to form specific PRC1 complexes [20]. These auxiliary proteins are capable of: (i) modulating the PRC1 catalytic activity, and (ii) targeting a specific site in the genome [24,25]. These complexes are named either canonical PRC1 (cPRC1) or non-canonical PRC1 (ncPRC1). The cPRC1 complexes arrange around either PCGF2 or PCGF4 and are constituted by one of the five chromodomain-containing paralogues (CBX2, 4, 6, 7 or 8), which are associated to the RAWUL domain of RING1, and a polyhomeotic subunit (PHC1, 2 or 3) [20,24]. On the other hand, the ncPRC1 complexes arrange around any of the six PCGF (PCGF1 to PCGF6), and they include RING1, as well as the intrinsically disordered RING1A and YY1-binding protein (RYBP, Uniprot: Q8N488) – or alternatively, its paralogue YY1-associated factor – which binds to the RAWUL domain of C-RING1B [26]. They also include various additional auxiliary proteins that are dependent on the type of PCGF protein present in the complex [25,27–29].

Citrullination is a PTM occurring in the presence of Ca^{2+} and catalyzed by L-arginine iminohydrolases (PADI), a family of proteins also known as peptidyl-arginine deiminases (EC 3.5.3.15). PADI4 (UniProt code: Q9UM07) is one of the five isoforms of L-arginine iminohydrolases (EC 3.5.3.15) found in humans, which convert arginine into citrulline, by deamination of its side-chain [30,31]. Among other cell locations, PADI4 is usually found in the cytoplasmic granules of neutrophils, eosinophils or macrophages [30]. PADI4 is also involved in the citrullination of histones H1 [30], H3 [32], H2A/H2B dimers [33] and H4 [34], and it is capable of counteracting histone arginine methylation [35]. We have recently reported that PADI4 binds to RYBP [36] and the RAWUL domain of RING1B, C-RING1B (residues 228–336 of the intact protein) [37], modifying some of the arginines in those polypeptides. In RYBP, the arginine post-translationally modified by PADI4 seems to be Arg53, although Arg56 is also very close, both being located at its N-terminus [36]; and, in C-RING1B, the modified arginine is Arg231 [37].

In this work, we wondered whether the isolated regions around those arginine residues of RYBP and C-RING1B polypeptides were also capable of binding to PADI4. To answer those questions, we synthesized two peptides for C-RING1B (the wild-type and the citrulline-modified one at Arg231), and three peptides for RYBP (the wild-type, the citrulline-modified at Arg53, and the citrulline-modified at Arg56 [36]). We first characterized the conformational features of these isolated peptides by diffusion ordered spectroscopy (DOSY), 1D- and 2D- ^1H NMR, and we found that all of them were disordered and monomeric in aqueous solution. To assess peptide binding to PADI4, we used fluorescence, isothermal titration calorimetry (ITC) and biolayer interferometry (BLI);

the three techniques showed binding for all peptides to PADI4. Furthermore, the wild-type peptides derived from C-RING1B and RYBP were citrullinated by PADI4 and, therefore, they were substrates, as observed by an enzymatic colorimetric assay (COLDER). Molecular simulations could reproduce the general trend observed in the experiments *in vitro*, and provided the additional information that the citrullinated peptides targeted the catalytic site of PADI4, as well. In general terms, we can conclude that the citrullinated peptides showed a slightly lower affinity for PADI4 than the corresponding wild-type ones.

2. Materials and methods

2.1. Materials

Imidazole, Trizma base, Kanamycin, DNase, 3-(trimethylsilyl) propionic acid-2,2,3,3- H_4 -sodium salt (TSP), SIGMAFAST protease inhibitor tablets, Ni^{2+} -resin, glycerol, N- α -Benzoyl-L-arginine ethyl ester (BAEE), L-citrulline, 3-hydroxyimino 2-butanone (diacetyl monoxime), thiosemicarbazide (TSC), SO_4H_2 (98 %), PO_4H_3 (85 %), $\text{NH}_4\text{Fe}(\text{SO}_4)_2 \times 12\text{H}_2\text{O}$, GSK484 (an inhibitor of PADI4), NaCl, antipyrine and ultra-pure dioxane were purchased from Sigma (Madrid, Spain). The β -mercaptoethanol was from BioRad (Madrid, Spain). Isopropyl- β -D-1-thiogalactopyranoside were obtained from Apollo Scientific (Stockport, UK). The detergent Triton X-100, the reducing agent TCEP (tris(2-carboxyethyl)phosphine), and the SDS protein marker (PAGEmark Tricolor) were from VWR (Barcelona, Spain). Amicon centrifugal devices with a molecular weight (MW) cut-off of 50 kDa were from Millipore (Barcelona, Spain). The rest of materials were of analytical grade. Water was deionized and purified on a Millipore system.

2.2. Protein expression and purification

The dimeric protein PADI4 (with 663 residues *per* monomer) was also purified as previously described [38]. PADI4 concentrations were determined by ultraviolet (UV) absorbance, employing an extinction coefficient at 280 nm estimated from the number of tyrosines and tryptophans (10 tryptophans and 13 tyrosines *per* monomer, $\epsilon_{280\text{nm}} = 73540 \text{ M}^{-1} \text{ cm}^{-1}$) [39].

2.3. Synthesis of the wild-type and citrullinated variants of peptides derived from C-RING1B and RYBP proteins

Based on previous results on how the C-RING1B and the intact RYBP were capable of binding to PADI4 [36,37], we synthesized five peptides, whose sequences are reported in Table 1. The peptides related to C-RING1B were: the wild-type (C-RING-wt), and the modified one with a citrulline at Arg231 (C-RING-R231). The RYBP-derived peptides were: the wild-type (RYBP-wt), the modified peptide with a citrulline at Arg53 (RYBP-R53), and the other one with a citrulline at position Arg56 (RYBP-R56), which is very close to the former position but less favored in the binding to PADI4 [36]. All the peptides were synthesized as acetylated and amidated, to avoid possible structural fraying at both termini. We also added in the RYBP-derived peptides a tyrosine at the C-terminal end to allow for the quantitative determination of their concentration in solution, by employing UV absorption at 280 nm [39]. The concentration of the C-RING1B peptides was determined from the absorbance at 280 nm of the sole phenylalanine present in their sequence ($\epsilon_{280\text{nm}} = 700 \text{ M}^{-1} \text{ cm}^{-1}$ [42]). All the peptides were produced by GenScript (Leiden, Netherlands), with a purity larger than 95 %.

2.4. Fluorescence

For the titration of PADI4 with the peptides, increasing concentrations of the latter, in the range 0–200 μM (for the C-RING1B peptides) and 0–40 μM (for the RYBP ones), were added to a solution with a fixed amount of the protein (2 μM in protomer units). Experiments were

Table 1
Hydrodynamic properties of the C-RING1B and RYBP peptides.^a

Polycomb peptides	D (cm ² s ⁻¹) (R_h , Å) ^b	R_h , Å ^c
D ²²¹ GASEIELVFRPHPTLMEKD ²⁴⁰ (C-RING-wt)	$(1.99 \pm 0.06) \times 10^{-6}$ (13 ± 1)	13 ± 2
D ²²¹ GASEIELVFCitPHPTLMEKD ²⁴⁰ (C-RING-R231)	$(1.90 \pm 0.03) \times 10^{-6}$ (14 ± 2)	13 ± 2
K ⁴⁸ GTSTRKPRINSQLVAQQ ⁶⁵ Y (RYBP-wt)	$(1.90 \pm 0.06) \times 10^{-6}$ (13 ± 2)	12 ± 2
K ⁴⁸ GTSTCItKPRINSQLVAQQ ⁶⁵ Y (RYBP-R53)	$(1.9 \pm 0.1) \times 10^{-6}$ (13 ± 3)	12 ± 2
K ⁴⁸ GTSTRKPCItINSQLVAQQ ⁶⁵ Y (RYBP-R56)	$(1.89 \pm 0.06) \times 10^{-6}$ (13 ± 2)	12 ± 2

^a All the peptides were amidated at their C-termini, and acetylated at their N-termini. The MW for the C-RING-wt peptide is 2284.57 Da. The MW for the RYBP-wt peptide is 2175.48 Da. The modified citrullines are indicated in bold.

^b The R_h was determined by using the translational diffusion coefficient of dioxane, R_h , of 2.12 Å [40] and the Stokes-Einstein equation. Errors in D values are fitting errors to an exponential curve and errors in R_h are due to propagation errors in the calculation [40].

^c Calculated from the scale law: $R_h = (0.027 \pm 0.01) \text{ MW}^{(0.50 \pm 0.01)}$ [41], where MW is the molecular weight of the peptide. Errors are due to propagation errors resulting from the equation used. The difference in MW between the wild-type species and each citrulline variant is 1 Da per introduced citrulline.

carried out at pH 8.0 (50 mM, Tris buffer) and 25 °C. The samples were excited at 280 and 295 nm, with the experimental set-up described in detail elsewhere [43]. In all cases, the appropriate blank corrections were subtracted. Spectra were also corrected for inner-filter effects during fluorescence excitation [44].

We attempted to calculate the dissociation constant for each PADI4/peptide complex, K_d , by fitting the binding isotherm, which was constructed by plotting the observed fluorescence change as a function of peptide concentration to the general binding model, explicitly considering peptide depletion due to the binding [45,46]:

$$F = F_0 + \frac{\Delta F_{\max}}{2[PADI4]_T} \left(([PADI4]_T + [Pep]_T + K_d) - \sqrt{((PADI4]_T + [Pep]_T + K_d)^2 - 4[PADI4]_T[Pep]_T)} \right) \quad (1)$$

where F is the measured fluorescence at any particular concentration of either the C-RING1B or RYBP peptides after subtraction of the corresponding blank; $\Delta F_{\max}$ is the largest change in the fluorescence of those peptides when they were forming the complex, taking as reference the fluorescence of each isolated chain; F_0 is the fluorescence intensity when no C-RING1B or RYBP peptide was added; $[Pep]_T$ is the total concentration of the corresponding C-RING1B or RYBP peptide, that was varied during the titration; and $[PADI4]_T$ is the constant concentration of PADI4. Each titration was repeated three times, using newly prepared samples. In all cases, the variations in the results were lower than 10 %. Fitting to Eq. (1) was carried out by using KaleidaGraph version 3.5 (Synergy software, Reading, PA, USA).

2.5. Nuclear Magnetic Resonance (NMR) spectroscopy

The NMR experiments on the C-RING1B or RYBP peptides were performed at 10 °C on a Bruker Avance 500 MHz spectrometer (Bruker GmbH, Karlsruhe, Germany), equipped with a triple resonance probe and z-pulse field gradients. Spectra were processed with Bruker TopSpin 2.1 (Bruker GmbH, Karlsruhe, Germany). All NMR experiments were carried out in 50 mM sodium phosphate buffer, at pH 7.2, in the presence of 50 µL of D₂O. Spectra were calibrated with TSP, by considering pH-dependent changes of its chemical shifts [47]; probe temperature was calibrated with methanol [47].

2.5.1. 1D-¹H NMR spectra

A number of 128 scans were acquired with 16 K acquisition points for

the homonuclear 1D-¹H NMR spectra of each peptide at ~1.2 mM concentration. Water signal was suppressed with the WATERGATE sequence [48]. The spectra were processed after zero-filling (with a total matrix size of 32 K), apodised with an exponential window and base-line corrected.

2.5.2. Translational diffusion NMR (DOSY)

The peptide concentrations in DOSY experiments [49] were ~100 µM, and 128 scans were performed, with a total amount of data of 16 K in the F₂ dimension. Measurements of the translational self-diffusion coefficient, D , were performed with the pulsed-gradient spin-echo sequence in the presence of 100 % D₂O. Details on the experimental conditions, calibration of the gradient strength and fitting of the resulting curves have been described elsewhere [40]. The estimated values of the hydrodynamic radii, R_h , of the peptides were determined from a comparison with the R_h of dioxane added to the solution [40].

2.5.3. 2D-¹H NMR spectra

Two-dimensional spectra of the C-RING1B and RYBP peptides were acquired in each dimension in phase-sensitive mode by using the time-proportional phase incrementation technique [50] and a spectral width of 5500 Hz; peptide concentration was the same used for the 1D-¹H NMR spectra. Standard TOCSY (with a mixing time of 80 ms) [51] and NOESY experiments (with a mixing time of 250 ms) [52] were performed by acquiring a data matrix size of 4096 × 512 points. The DIPSI (decoupling in the presence of scalar interactions) spin-lock sequence [53] was used in the TOCSY experiments with a relaxation time of 1 s. A number of 96 scans were acquired per increment in the first dimension. NOESY spectra were collected with 128 scans per increment in the first dimension with a relaxation time of 1 s. In both spectra, residual water signal was removed by using the WATERGATE sequence [48]. Data were zero-filled, resolution-enhanced with a square sine-bell window function optimized in each spectrum, and baseline-corrected. The ¹H resonances were assigned by using the standard sequential assignment method [54]. The chemical shift values of H_α protons in random-coils were calculated by considering the neighboring residue effects [54–56].

2.6. Isothermal Titration Calorimetry (ITC)

The binding affinities for the RING and RYBP peptides interacting with PADI4 were determined in a PEAQ-ITC titration calorimeter (MicroCal, Malvern-Panalytical, Malvern, UK). The titrations were carried out in buffer 50 mM Tris at pH 8.0, at a temperature of 25 °C. Protein solutions in the cell at 5–20 µM were titrated with each of the peptide solutions at 50–200 µM from the injection syringe. In each experiment a sequence of nineteen injections of 2 µL of titrant solution spaced evenly over 150 s was programmed, with a stirring speed of 750 rpm and a reference power of 10 µcal/s. Experiments were repeated twice. Additional experiments employing GSK484, a well-known orthosteric PADI4 inhibitor, were performed to assess whether the peptides compete for the binding to the active site. In this case, GSK484 was added to the calorimetric cell solution at a concentration of 500 µM.

The binding isotherms were obtained by integrating the heats per injections in the thermograms and normalizing by the amount of ligand injected per injection. Data analysis was performed by non-linear least-squares regression data analysis employing a user-defined fitting routine in Origin 7.0 (OriginLab, Northampton, MA) and considering a single binding-site model. The equilibrium association constant, K_a (and the dissociation constant, $K_d = 1/K_a$), and the apparent binding stoichiometry, n , were estimated.

2.7. Verification of citrulline formation by a colorimetric assay

A photometric method was carried out to determine whether PADI4

was capable of citrullinating BAEE (as a control) or, alternatively, either the C-RING-wt or RYBP-wt peptides. This method is based in the reaction of citrulline with oximes in strong acids in the presence of TSC. The protocol to prepare the colour developed reagent (COLDER) and the corresponding tests have been described elsewhere [57–59] and applied to detect the enzymatic activity of PADI4 under different conditions [60–62]. The protocols and the corresponding controls for peptides susceptible of being citrullinated have been extensively described elsewhere with other peptides from different proteins [61,62]. Concentrations of both wild-type Polycomb peptides were 100 μ M and that of PADI4 was 0.5 μ M.

2.8. Biolayer Interferometry (BLI)

2.8.1. Experimental design

The association (k_{on}) and dissociation (k_{off}) rate constants of the binding reactions of the Polycomb peptides to PADI4 were determined by using a BLItz system (ForteBio, Pall, Barcelona, Spain) [63–67]. The buffer used in the experiments was that recommended by the manufacturer (Octet kinetics Buffer 10 $\times$ (10 $\times$ KB), PN 18-1105, Sartorius). As PADI4 had a His-tag, it was immobilized on His-tag biosensors (Octet NTA Biosensors, PN 18-5101) at 0.3 μ M (in protomer units). Before the loading, each Polycomb peptide was tested for binding at the largest concentration assayed against unloaded tips; in all the cases, we did not observe any binding. The C-RING1B and RYBP peptide concentrations were in the range from 20 to 120 μ M during the association step. The general scheme of the protein-association/dissociation reactions was: 30 s of initial baseline acquisition with the 10 $\times$ kinetics buffer; 120 s of loading PADI4 into the His-tag biosensor; 30 s of baseline acquisition with the 10 $\times$ kinetics buffer; 120 s of association of each Polycomb peptide to the biosensor-immobilized-PADI4; and 120 s of dissociation of the corresponding peptide from the biosensor-immobilized-PADI4. Baseline measurements in the presence of unloaded tips were subtracted from the measurement taken with the loaded tip. Experiments were repeated twice with new biosensors for each of the peptides.

2.8.2. Fitting of the sensorgrams

Fitting of the sensorgrams was carried out as described elsewhere [61,62,67]. Briefly, the response during the association step, $R(t)$, to the biosensor-bound-PADI4, was fitted to:

$$R(t) = R_{eq} - R_{eq} e^{(-k_{obs}(t-t_0))} - \dot{R}_{eq}(t - t_0) \quad (2)$$

where the last term was included to consider the linear drift observed at longer times. In this equation, R_{eq} is the steady-state (or equilibrium) response obtained at infinite time, when $dR(t)/dt = 0$, and $t_0 = 180$ s is the time at which the association step started. The value of k_{obs} obtained at different peptide concentrations, $[Pep]$, is given by the pseudo-first-order equation:

$$k_{obs} = k_{on} [Pep] + k_{off} \quad (3)$$

where the k_{on} and k_{off} are, respectively, the kinetic association and dissociation rates of the binding reaction of the corresponding peptide to PADI4.

The dissociation process was fitted with $R(t)$ given by:

$$R(t) = R_2 - R_1 e^{(-k_{off}(t-t_0))} - \dot{R}_{eq}(t - t_0) \quad (4)$$

where $t_0 = 300$ is the time at which the dissociation of the peptide from the biosensor-bound-PADI4 started in our experimental set-up, R_1 is the response level when dissociation starts, R_2 is the response at infinite time values, and the last term, $\dot{R}_{eq}$, was included to consider the linear drift observed at longer times. Separated fittings of the association and the dissociation steps to Eqs. (2) and (4), at the different peptide concentrations, were carried out by using KaleidaGraph. Fitting to Eq. (3) (the

so-called pseudo-first-order plot) was also carried out by using KaleidaGraph.

2.9. Molecular docking simulations

Molecular docking simulations were carried out to study the binding locations of the C-RING1B and RYBP peptides on PADI4. The peptides were built by using UCSF Chimera [68], with acetylated and methylated capping at their N and C terminus, respectively. The structure of PADI4 was built starting from its crystallographic conformation (PDB entry: 3APN [69]), and missing loop portions were modeled as previously described [38]. To reduce the computational cost, a single protein monomer (663 residues) of PADI4 was used in the docking experiments. The software PyMol, version 2.1.1 (<https://www.pymol.org>) was used to display the docking poses.

The large number of rotational degrees of freedom of the peptides (78 rotatable bonds for C-RING-wt and its citrulline-modified counterpart, and 82 for RYBP-wt and its two citrullinated ones) precluded the use of many long-established docking algorithm (e.g., 32 rotatable bonds is the limit considered reliable for AutoDock Vina [70]). For this reason, the generative diffusion model DiffDock was used [71], which uses a three-layer graph neural network to score potential docking poses and rank them according to a confidence value. To allow for a direct comparison with the binding score of other peptides we previously proved to bind PADI4 in experiments and simulations [61,62,72], default docking parameters were used for DiffDock: 20 diffusion time divisions, 18 actual inference steps, and no final step noise. Ten simulation runs were performed and, among the top-ranked docking poses, the five most favorable ones were considered.

Additional runs were made to reduce the computational complexity of the simulations, by considering: (i) the peptides shortened by three residues on each termini (resulting in 54 rotatable bonds for each of the five peptides); and (ii) the sole PADI4 domain where docking poses tended to accumulate (residue 295–663 of the protein monomer).

3. Results

3.1. Conformational features of the C-RING1B and RYBP peptides

To perform the characterization of the conformational features of the C-RING1B and RYBP peptides (Table 1), we carried out 1D- and 2D- 1 H NMR studies. We did not use fluorescence to characterize the structural features of the isolated peptides, since: (i) to allow for measurement of their concentration, the RYBP peptides had a non-natural tyrosine at the C-terminal region, whose maximum wavelength in the fluorescence spectrum does not change in the presence of well-defined or rigid structure [42]; and (ii) the C-RING1B ones had a sole phenylalanine.

The C-RING1B and RYBP peptides were monomeric, as concluded from: (i) the value of D measured by the DOSY experiment (Table 1); and (ii) the estimated R_h obtained from the comparison with the experimentally measured D of dioxane by using the Stokes-Einstein relationship [40]. In fact, the estimated values of R_h were similar to those obtained theoretically for random-coil polypeptides [41] with the same MWs (Table 1).

The disordered character of the Polycomb peptides was evident from the 1D- 1 H NMR spectra (Fig. 1), which showed a clustering of the signals of all the methyl protons of Leu, Ile and Val residues at ~ 0.8 –1.0 ppm [54]. Similarly, the amide protons were clustered between 8.0 and 8.6 ppm in all peptides (Fig. S1). These values for the methyl and amide protons are typical of residues belonging to disordered chains [54].

To further confirm the disordered nature of the C-RING1B and RYBP peptides, we also carried out 2D- 1 H NMR experiments. The peptides were found to be mainly disordered in solution, as suggested by two pieces of evidence. First, the sequence-corrected conformational shifts ($\Delta\delta$) of H_α protons of all peptides for unambiguously assigned residues were within the commonly accepted range for random-coil peptides

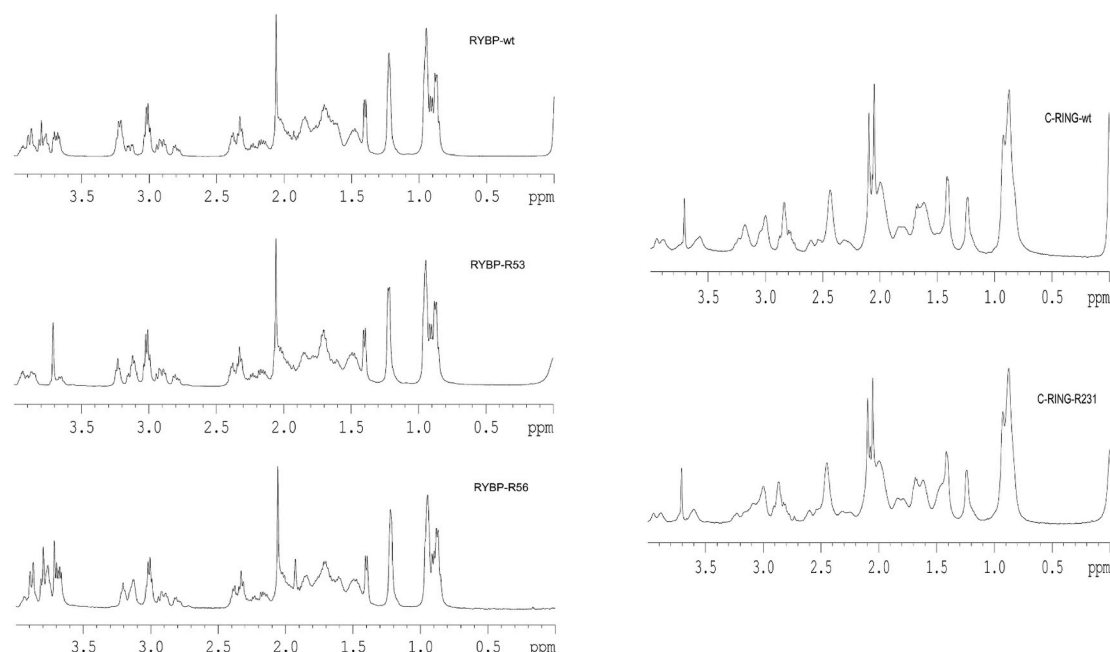


Fig. 1. Conformational features of the isolated Polycomb peptides in solution as measured by 1D- ^1H -NMR: Methyl regions of the 1D- ^1H NMR spectra of the isolated C-RING1B and RYBP peptides. The signal appearing at 0 ppm in some of the regions is that of TSP. Spectra were acquired at 10 °C and pH 7.2 (50 mM, sodium phosphate buffer).

($\Delta\delta \leq 0.1$ ppm) (Tables ST1 to ST5). And second, no long- or medium-range NOEs were detected, but only sequential ones (i.e., $\alpha\text{N}(i, i+1)$ and $\beta\text{N}(i, i+1)$) in the polypeptide patches fully assigned (Fig. 2). It is interesting to note that citrullination did not affect the chemical shifts of backbone or side-chain protons (Tables ST1 to ST5).

It is important to indicate that the region around Arg231 in C-RING1B appears disordered in the X-ray structure of this domain (PDB code: 3H8H [73]). Furthermore, the region around Arg53 and Arg56 in RYBP in the cryo-electron microscopy structure of a complex with other proteins show that both arginines also belong to a disordered

polypeptide patch (PDB code: 8PP6 [15]). Therefore, the conformation of the isolated peptides obtained in our experiments was no more ordered than the one they have in the intact polypeptide chains.

Then, we can conclude that all isolated C-RING1B and RYBP peptides in solution were monomeric and disordered.

3.2. C-RING1B and RYBP peptides were capable of binding to PADI4

We carried out fluorescence titrations to determine the affinity constant of the peptides towards PADI4, fixing the concentration of the

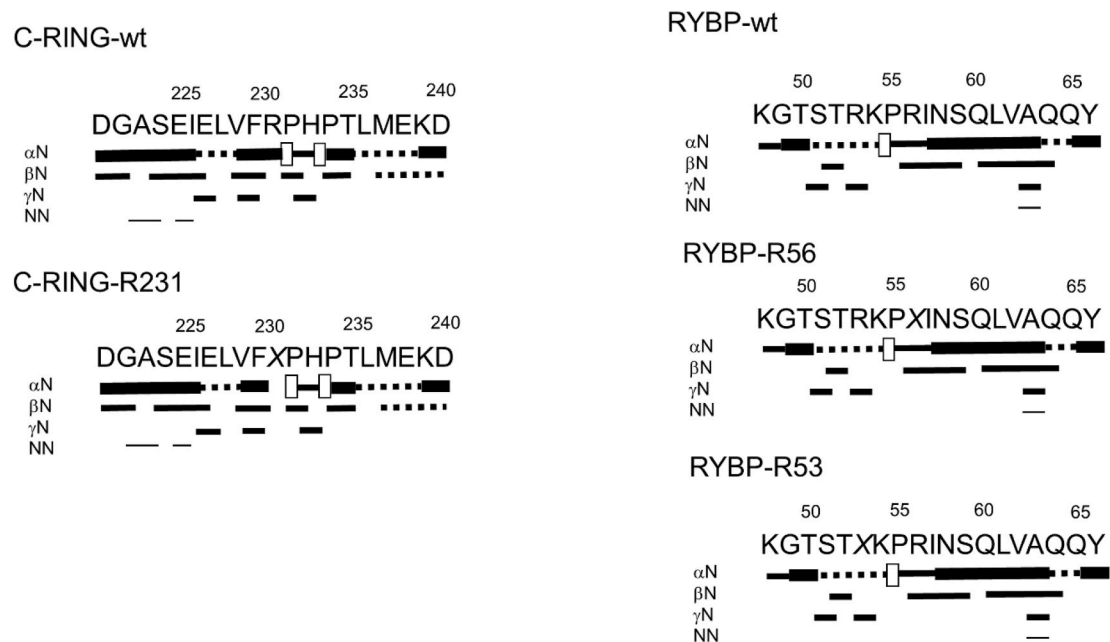


Fig. 2. NOE diagrams of the isolated C-RING1B and RYBP peptides in aqueous solution from the 2D- ^1H NMR NOESY spectra: NOEs are classified as strong, medium or weak, and represented by the height of the bar underneath the sequence; signal intensity was judged from the NOESY experiments. The dotted lines indicate NOE contacts that could not be unambiguously assigned due to signal overlapping. A white square indicates an $\alpha\delta(i, i+1)$ NOEs involving a proline residue.

enzyme, and raising that of the corresponding peptide. We also used ITC to determine the affinity constant. And, finally, we used BLI to determine the kinetic parameters of the binding reactions, and to estimate an affinity constant to compare with those obtained by the other two techniques, by assuming a two-state process for the binding of each peptide to the enzyme. It is important to pinpoint that, in the context of the fluorescence experiments, we are assuming that the unbound and the free species were the only ones significantly populated (Eq. (1)). However, in the context of the BLI experiments, if we assume that there are more than those two species (that is, an intermediate state may exist between the free and bound states) the equilibrium dissociation constant cannot be obtained as $K_d = k_{off}/k_{on}$, but rather by using more complex relationships [74].

We first carried out fluorescence titrations with each of the peptides and PADI4. We observed a decrease of the fluorescence intensity as the concentration of the peptide was increased (by following the titration at either 280 or 295 nm) at any of the explored emission wavelengths. In this way, we were able to determine a K_d value for each peptide (Fig. 3 and S2, and Table 2). In the case of C-RING1B peptides, as they only contained a phenylalanine residue, the variation in the fluorescence intensity indicated that some of the tryptophans (10) and/or tyrosines (13) present in PADI4 were involved in the binding to the peptide. For the RYBP peptides, since they only contained an additional tyrosine at their corresponding C-terminal region, and variations in fluorescence intensity were also observed during the titration, then, these results suggested that some of the tryptophan residues of PADI4 were involved in the binding to these peptides.

The calorimetric titrations of the Polycomb peptides provided direct evidence for their interaction with PADI4 (Table 3, Fig. 4). The binding affinity of the unmodified C-RING1B and RYBP peptides was in the low micromolar range, similar to other previously characterized non-citrullinated peptides [61,62,72]. The citrullination of particular arginines caused a decrease in the binding affinity; interestingly, the most significant change was observed in the RYBP-R56 peptide. Additional experiments performed by adding GSK484, an orthosteric inhibitor of PADI4, indicated that occupation of the active site in PADI4 precluded the interaction with the wild-type peptides (Fig. S3); therefore, the Polycomb wild-type peptides must bind to the active site of PADI4. Although this piece of evidence by itself cannot rule out the possibility of GSK484 binding to PADI4 and inducing a conformational change that hindered the interaction with the peptide in an allosteric fashion, the fact that wild-type peptides could be citrullinated by PADI4 confirmed the hypothesis that they bind to the active site (see below in this section).

Finally, we carried out BLI experiments to have another independent estimate of the affinity of the binding of the peptides to PADI4 (Table 4, Fig. 5 and S4). It is important to indicate that due to the immobilization of PADI4 to the biosensor, some of the potential peptide-binding regions in the enzyme molecules could get occluded, hampering the peptide

Table 2

Binding affinities of C-RING1B and RYBP peptides to PADI4 as measured by intrinsic fluorescence.^a

Peptide	K_d (μ M) ^c
C-RING-wt	13 $\pm$ 8
C-RING-R231	23 $\pm$ 18
RYBP-wt	23 $\pm$ 20
RYBP-R53	8 $\pm$ 5
RYBP-R56	14 $\pm$ 10

^a Determined at 25 °C.

Table 3

Binding affinities of C-RING1B and RYBP peptides interacting with PADI4 as determined by ITC.^a

Polycomb peptides	K_a (M^{-1})	K_d (μ M)	n
C-RING-wt	$4.5 \cdot 10^5$	2.2	0.9
C-RING-R231	$1.1 \cdot 10^5$	8.8	1.0
RYBP-wt	$3.4 \cdot 10^5$	2.9	0.9
RYBP-R53	$7.9 \cdot 10^4$	13	1.0
RYBP-R56	$2.6 \cdot 10^4$	39	1.0

^a Relative error in K_a and K_d is 30 %, obtained from two measurements. Determined at 25 °C.

binding. Although we acknowledge such limitation that constitutes a well-known drawback of this technique [63–66], in our case we wanted to have an additional method to further confirm the binding of the peptides to the enzyme together with fluorescence (which may not yield reliable affinity constants when there is a lack of structural changes around a tryptophan and/or tyrosine) or ITC (whose results may be difficult to interpret when there is a lack of a significant heat exchanged). In general terms, for both types of the Polycomb peptides, the values of the k_{off} rates were larger when citrulline was present, compared to those for the wild-type species (Table 4). These findings resulted in larger dissociation constants for the citrullinated species from both Polycomb peptides than for the wild-type ones. Furthermore, in the case of the RYBP peptide citrullinated at position 56, the value of k_{on} was much smaller (one order of magnitude) than for the peptide citrullinated at position 53 (Table 4), even though the value of the k_{off} was similar for both modified peptides (~ 0.50 s⁻¹). This suggests that, from a kinetic point of view, the binding of a modified arginine at position 56 is disfavored with respect to modification at position 53.

Then, all in all, although the particular PADI4-peptide binding affinities determined by the different techniques somewhat differ, they showed a common trend: dissociation constants in the low micromolar range for the unmodified peptides and larger dissociation constants for the citrullinated species, in both Polycomb peptides, compared to the wild-type ones.

Since the affinity for PADI4 of both wild-type peptides were the

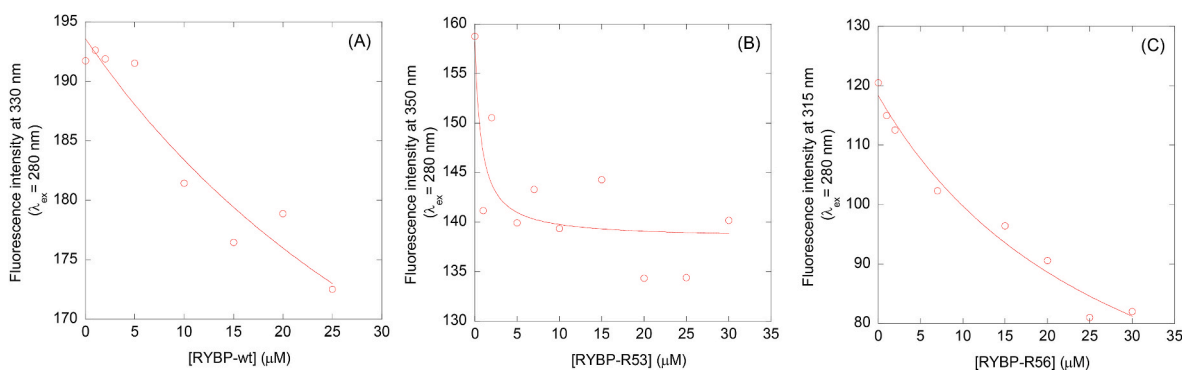


Fig. 3. Fluorescence titration curves of RYBP peptides to PADI4: Titration curves of the three RYBP peptides with the enzyme obtained by excitation at 280 nm, observed at different wavelengths. All experiments were performed at 25 °C. Experiments were repeated three times with new samples.

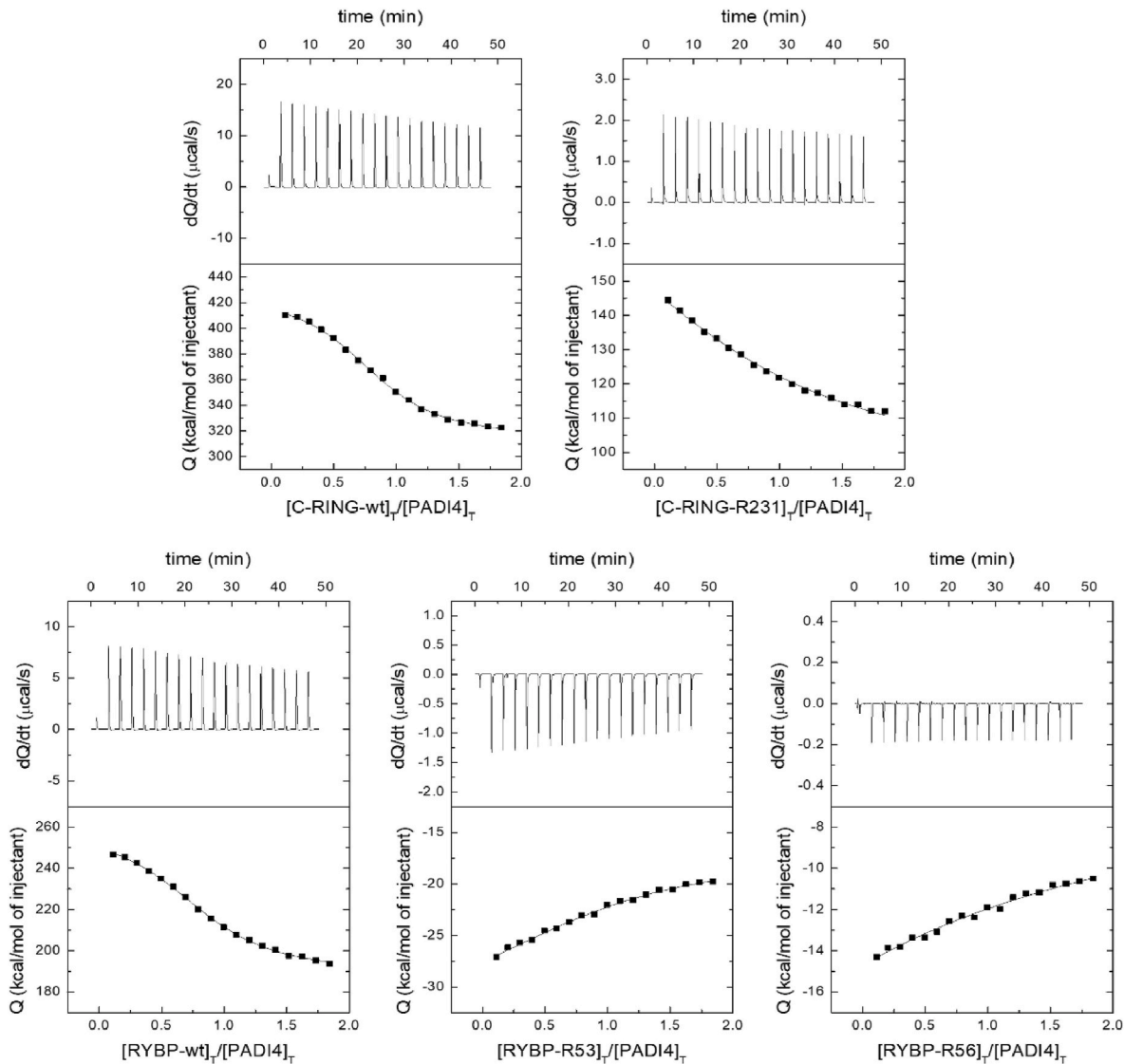


Fig. 4. Binding of Polycomb peptides to PADI4 as monitored by ITC: Calorimetric titrations of PADI4 with the five C-RING1B and RYBP peptides. Upper panels show the thermograms (thermal power as a function of time) and lower panels show the binding isotherms (ligand-normalized heat effects *per* injection as a function of the molar ratio in the calorimetric cell). Continuous curves correspond to the non-linear least-squares fittings according to a single ligand binding site interaction model. Experiments were carried out at 25 °C. Experiments were repeated twice.

Table 4
Kinetic parameters of the binding of the Polycomb peptides to PADI4 as determined by BLI.

Polycomb peptides	k_{on} ($\mu M^{-1} s^{-1}$) ^a	k_{off} (s^{-1})	K_d (μM) ($= k_{off}/k_{on}$) ^b
C-RING-wt	0.0020 ± 0.0002	0.053 ± 0.013	26 ± 10
C-RING-R231	0.0012 ± 0.0002	0.14 ± 0.01	114 ± 40
RYBP-wt	0.021 ± 0.005	0.11 ± 0.21	5.5 ± 5.0
RYBP-R53	0.021 ± 0.004	0.49 ± 0.14	23 ± 10
RYBP-R56	0.008 ± 0.003	0.53 ± 0.12	66 ± 35

^a Obtained from pseudo-first order plots. Error for k_{on} is the fitting error to Eq. (3). Determined at 25 °C.
^b Errors are obtained from propagation errors for k_{on} and k_{off} . The estimation of the K_d is carried out assuming that the binding process was two-state [74].

largest among all the peptides studied, either measured by ITC (Table 3) or BLI (Table 4), and they do not contain any citrulline residue, we tested whether such peptides were capable of being citrullinated by PADI4 by using a colorimetric assay. We observed that C-RING-wt and RYBP-wt were substrates of the protein, since they were citrullinated in the

presence of PADI4 (Fig. S5) in agreement with the ITC results in the presence of GSK484 (Fig. S3) (i.e., the peptides undergo citrullination by PADI4 and their binding is blocked by an orthosteric competitive inhibitor). Therefore, we could conclude that this finding is a direct proof that both wild-type peptides targeted the active site of the enzyme.

3.3. All polycomb peptides targeted the catalytic active site of PADI4

Whereas the colorimetric assays provided a proof of the binding to the catalytic active site for the two wild-type Polycomb peptides (i.e., RING1B-wt and RYBP-wt), for the citrulline-modified peptides we lacked an analogous evidence. The presence of an allosteric rather than orthosteric binding site might be suspected in this case, because dissociation of a citrullinated site occurs naturally after catalysis; therefore, affinity towards the active site could be comparatively less favorable than in the presence of a native arginine. With the aim of getting further details on the binding propensity of our Polycomb peptides towards PADI4, we performed molecular docking simulations by using the diffusion-based algorithm DiffDock [71].

Fig. 6 and S6 show the most favorable binding modes obtained by

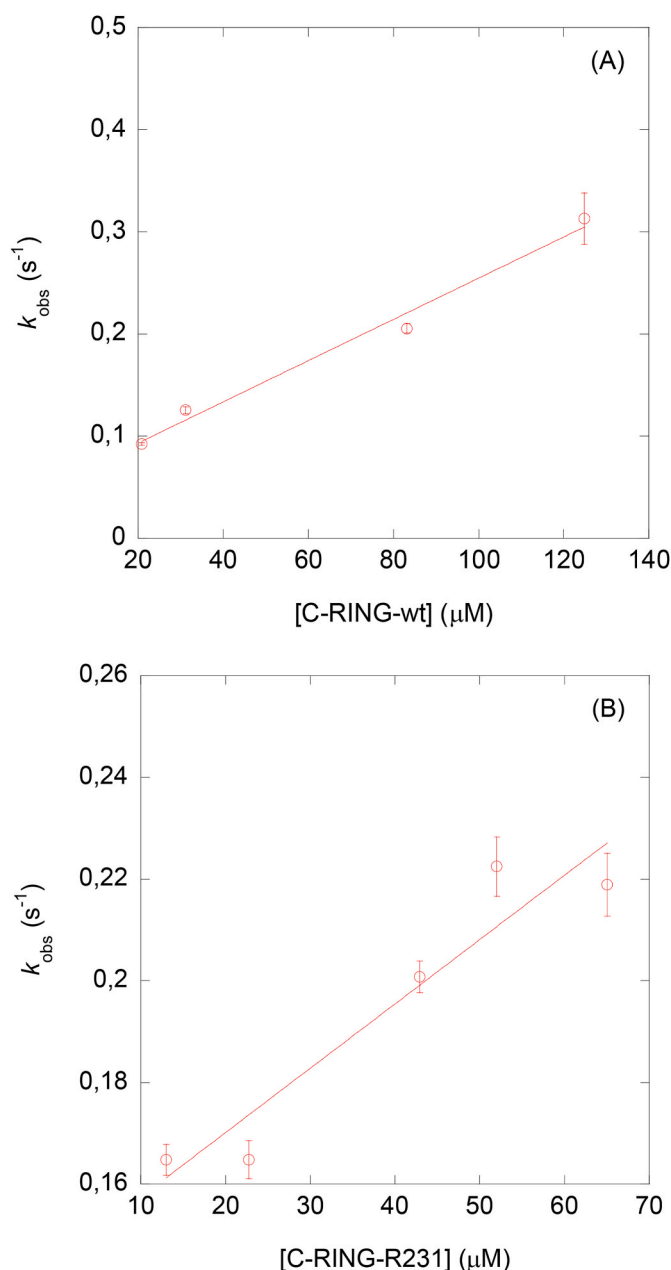


Fig. 5. Binding of the C-RING1B peptides to PADI4 as monitored by BLI: Pseudo-first-order plots of the binding of C-RING-wt and C-RING-R231 peptides to PADI4. The error bars of the fittings to Eq. (2) at each peptide concentration are indicated. Experiments were carried out at 25°C. Experiments were repeated twice.

blind docking of our peptides on a single monomer of PADI4. While docking poses with a lower rank could be found occasionally on other spurious locations (not shown), the best docking poses for all wild-type and citrulline-modified peptides derived from C-RING1B and RYBP targeted the active site of PADI4. In particular, the key residue Cys645, that performs the catalytic reaction from arginine to citrulline, was in contact with all the binding modes for all the peptides. The binding region of PADI4 included a few nearby tryptophans (e.g., Trp347, located within 1 nm from Cys645), which could contribute to the variations in the fluorescence signal observed upon binding (Fig. 3).

Table 5 reports the confidence scores of DiffDock (i.e., the dimensionless values with arbitrary units used by the algorithm to score the docking poses) found for each peptide. These values were approximately in the range from -3.0 to -3.5 , which is markedly less favorable than

the confidence score ($c \approx -1.5$) we have formerly reported for lipids binding to PADI4 [75], or that could be expected in general for the association of drug-like compounds to proteins [71]. Nevertheless, similar or even less favorable results were previously obtained for other peptides experimentally proved to bind to PADI4, including both wild-type and citrullinated species. In fact, confidence scores of -3.7 ± 0.3 were found for 23-residue peptides mimicking the nuclear localization signal of Inhibitor of Growth 4 [62], and values in between 3.5 and 4.0 for shorter citrullination-prone peptides (15 or 16 residues long) derived from the ubiquitin ligase MDM2 [61].

To decrease computational costs, additional simulations were performed by using shortened peptides missing three residues at each end, and employing as docking host the C-terminal PADI4 domain (residue 295–663) where docking poses accumulated. The confidence scores were in this case more favorable (Table S6), about -2.6 ± 0.2 , and again with no clear difference between wild-type and citrullinated peptides. The catalytic protein residue Cys645 was the target of all the peptides, including in some cases direct contact with the arginine side chain. (Fig. S7). In particular, for RYBP peptides, contacts with at least one arginine (for RYBP-wt) or a competition between the arginine and citrulline (for RYBP-R53 and RYBP-R56 peptides) was observed.

4. Discussion

We have previously shown that the C-RING1B and RYBP intact polypeptides were capable of binding to PADI4, with affinity constants in the order of 10 μM (from the fluorescence measurements) and 1.5 μM (from the point of view of ITC) [36,37]. In this work, we showed that the isolated regions of each of those proteins, which are citrullinated by PADI4, were also capable of binding to PADI4 with affinity constants similar to those of the corresponding parental proteins, as obtained by fluorescence (Table 2) and ITC (Table 3). Furthermore, as it happens with the intact polypeptides, our colorimetric assays provided a direct proof that the wild-type peptides were substrates of the enzymes (Fig. S5); it is interesting to note that ITC experiments in the presence of GSK484 (an inhibitor of PADI4) also suggested that the peptides aimed to the active site. These results suggest that the isolated peptides could be used as competitors of the interactions of the corresponding parental proteins with PADI4. Furthermore, keeping in mind that the isolated peptides could have different behavior to that of the corresponding polypeptide region within the intact proteins, they might be used as the starting point to obtain candidate binders that could work as inhibitors of PADI4, and perhaps also other members of the family of PADI proteins. In the case of the Polycomb peptides, based on the enzymatic results (Fig. S5), we showed that the selected polypeptides were substrates of PADI4.

Although from the point of view of fluorescence the apparent affinity constant measured are not very different from each other (Table 2), the affinity constants measured by ITC (Table 3) and those estimated from BLI (Table 4) indicate that the affinities of the citrullinated peptide were smaller than those of the corresponding counterparts (i.e., larger K_d values). These findings about how citrullination affects binding to PADI4 agree with those previously found for other citrullinated peptides with no relationship to RYBP or RING1B [61,62,72], and suggest that once the substrate protein is citrullinated, PADI4 tends to dissociate from the post-translationally modified protein, which is a phenomenon observed in other enzymes for their substrate and the products of the reaction [74]. Furthermore, this dissociation of the modified protein occurs much faster than for the wild-type peptides (Table 4), which agrees, in broader terms, with the kinetic parameters of the dissociation reaction also measured for other citrullinated peptides [61,62,72].

It is interesting to note that, in a previous work of ours, the Arg53 of RYBP was predicted to be the residue prone to citrullinate [36]; however, the close proximity of both Arg53 and Arg56 in our structural model of the corresponding RYBP region bound to the catalytic residue Cys645 of PADI4 prompted us to study the two citrullinated peptides. In

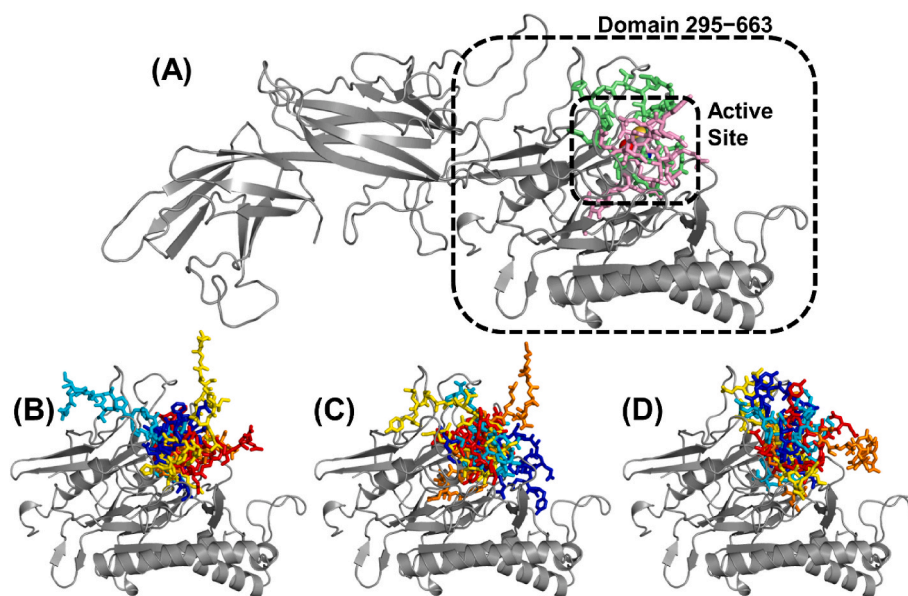


Fig. 6. Docking poses of C-RING1B and RYBP peptides on PADI4 obtained by using the algorithm DiffDock: (A) Best docking poses for wild-type peptides, C-RING-wt (green) and RYBP-wt (pink). PADI4 monomer in ribbon representation, with the C-terminal domain (residues 295–663) and active site (around residue Cys645) explicitly labeled. The best five docking poses (red, orange, yellow, cyan, and blue) of citrullinated peptides are shown for (B) C-RING-R231, (C) RYBP-R53, and (D) RYBP-R56.

Table 5

Confidence score for the five most favorable docking poses of C-RING1B and RYBP peptides to PADI4 as obtained with the DiffDock algorithm.

Peptide	Confidence score
C-RING-wt	-3.13 ± 0.26
C-RING-R231	-3.31 ± 0.12
RYBP-wt	-3.19 ± 0.15
RYBP-R53	-3.29 ± 0.24
RYBP-R56	-3.35 ± 0.25

fact, both arginines interact with the nucleosome acidic patch during the read-write mechanisms of the RYBP-PRC1 complexes [14,15], forming an R-ring-acidic patch interface. Our current results showed that the affinity for PADI4 was smaller for the peptide with citrullination occurring at Arg56 than that for the peptide modified at Arg53, either as measured by ITC (Table 3) or BLI (Table 4). The latter also suggested that the velocity of separation from PADI4 was much larger for the modification occurring at Arg56. Thus, citrullination at Arg56 appeared to be less favored from both a thermodynamic and kinetic point of view.

We also investigated whether the citrullinated peptides, despite their smaller affinity for PADI4, could be also capable of binding to PADI4 and aiming at its active site, similarly to the wild-type polypeptides. Our molecular docking results, obtained by using a blind simulation protocol, showed that the target hot-spot of all the peptides was the catalytic site of PADI4 rather than an allosteric site. Although the confidence scores of the docking poses were moderate, the values found were more favorable than those we had recently reported for other wild-type and citrullinated peptides of comparable size, all of them docking to PADI4 [61,62,72]. The confidence scores obtained for the two C-RING1B and RYBP wild-type peptides were similar to those obtained for the citrullinated ones (Tables 5 and S5). This indicates that the simulation does not have a sufficient sensitivity to detect any variations due to the small difference in the peptides (citrulline and arginine differ by a single atom), similarly to fluorescence (Table 2), and at variance with both ITC (Table 3) and BLI (Table 4).

Consistently with the stochasticity of the diffusion-based docking algorithm used and the variability observed in the results of different

simulation runs, we interpret the docking poses obtained as late-stage intermediates along the PADI4/peptide binding process rather than ‘ideal’ binding configurations (i.e., they were not equivalent to structures that might be obtained by crystallography or cryo-electron microscopy). Furthermore, as previously hypothesized [61,62], such variability may suggest an induced-fit association mechanism for the corresponding Polycomb protein regions, which would limit the conformational landscape in the process of recognition and binding to PADI4.

We pinpoint that in our previous works, we have shown that the interaction between RING1B and PADI4 [37], and between RYBP and PADI4 [36], was occurring in different cancer cell lines. We do not know yet whether such interactions, and citrullination of the corresponding arginines (Arg231 for C-RING1B, and Arg53 or Arg56 for RYBP), are important for cancer development, but the fact that the corresponding wild-type peptides bind to the active site of the enzyme suggest that these peptides, once they are properly delivered into the cancer lines, could be used as potential inhibitors of such interactions.

5. Conclusions

To sum up, we have shown that isolated peptides encompassing the binding site of two proteins belonging to the Polycomb system, RING1B and RYBP, were also substrates of the enzyme PADI4. Our study may constitute one more step towards the understanding of complex epigenetic mechanisms, because the Polycomb system intervenes in crucial pathways that include development during embryogenesis, preservation of cell plasticity for specialization, and safeguarding of cell identity during growth. All these mechanisms are crucial in preventing misregulation or mutations that might lead to cancer. The effect of citrullination in this context is still poorly understood, and studies on this irreversible PTM seem to be still too sparse to get a general picture. In this context, our study constitutes one more indication of involvement of PADI4 (and possibly other PADI isoenzymes) on physiopathological process involving Polycomb proteins. Although it might be too early to draw conclusions on all the possible implications, we hope that this could have an impact to reveal future opportunities in cancer therapies.

Supporting information available

Seven figures showing: the amide region of the 1D-¹H NMR spectra of the Polycomb peptides (Fig. S1); fluorescence titrations for some of the Polycomb peptides (Fig. S2); the ITC thermograms of the binding of C-RING-wt and RYBP-wt peptides to PADI4 in the presence of the competing inhibitor GSK484 (Fig. S3); the pseudo-first order plots for the binding of RYBP peptides (Fig. S4); the results of the COLDER assays with C-RING-wt and RYBP-wt peptides (Fig. S5); the most favorable docking poses of C-RING-wt and RYBP-wt peptides on PADI4 obtained by using the algorithm DiffDock (Fig. S6); and interactions between selected residues of PADI4 and arginine/citrulline residues of the peptides in simulated docking poses (Fig. S7). Six tables containing the ¹H NMR assignments of the Polycomb peptides (Tables ST1 to ST5) and a Table (ST6) containing the confidence scores for the five most favorable docking poses of shortened C-RING1B and RYBP peptides to the C-terminal domain of PADI4 as obtained with the DiffDock algorithm.

CRedit authorship contribution statement

María Gabriela Álvarez-Rodríguez: Formal analysis, Investigation, Methodology, Writing – review & editing. **Emanuel Capone:** Formal analysis, Investigation, Writing – review & editing. **Sonia Vega:** Formal analysis, Investigation, Writing – review & editing. **Virginia Rejas:** Investigation, Methodology. **Bruno Rizzuti:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Adrian Velazquez-Campoy:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **José L. Neira:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.abb.2026.110955>.

Data availability

Data will be made available on request.

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